



VISIBLE EMISSIONS OBSERVATIONS

In the late 1800's and the early 1900's, visible emissions were treated as a public nuisance.



**1850s
to
1930s**

1859

First court case regarding emissions in New Orleans.

1881

First cities to adopt smoke control ordinances: Chicago & Cincinnati

1897

U.S. begins use of Ringelmann Scale.

1910

Rochester, NY upholds Ringelmann, and creates smoke restrictions.

1912

23 of the 28 largest US cities have smoke ordinances.

1916

Northwestern Laundry vs City of Des Moines.



By the 1940's, smoke and air pollutants were recognized as detrimental to human health.



**1940s
to
1960s**

1945

Los Angeles develops equivalent opacity concept.

1948

Surgeon General declared smoke and pollutants health hazards.

1950

CA Rule 50A limits emissions. Rule used widely in other states.

1955

First federal action: Federal Air Pollution Act.

1963

Initial Clean Air Act. Provided funding to agencies.

1968

Joint industry & government study AP-30.



The EPA is formed. By the end of the 1980's, Method 9 corrected excessive emissions at most facilities.



**1970s
to
1990s**

1970

Environmental Protection Agency (EPA) formed.

1970

Clean Air Act is updated.

1974

EPA promulgates Method 9.

1977

Amendments to Clean Air Act.

1982

EPA promulgates Method 22.

1990

Clean Air Act updated to include Title V.



In 1997, Title V began to shift focus from enforcement to maintenance of air quality through continued use of Method 9.